

Project Executable Files

Date	02 October 2026
Team ID	SWTID-2026-5625
Project Name	FitBuddy – AI Fitness Plan Generator Using Gemini Models
Maximum Marks	

Team Members and Roles

1. Sharmila D – Team Lead

- Managed the overall project development.
- Handled project pre-requisites and topic links.
- Worked on the core functionalities of FitBuddy.
- Coordinated the team members and project activities.
- Prepared the application for local deployment.

2. Bhavana A – Backend Developer

- Worked on the project workflow.
- Developed and handled the FastAPI backend.
- Worked on routing and user-input handling.
- Tested the application during local deployment.

3. Rajashree K – AI/GenAI Developer

- Worked on Gemini AI integration.
- Researched the suitable Gemini model for the project.
- Worked on the main application logic.
- Contributed to the conclusion and AI-related functionality.

4. Siva Sundari V – Frontend/UI Developer

- Worked on the application architecture.
- Designed the user interface.
- Developed the frontend components.
- Focused on creating a simple and user-friendly interface.

5. Praveena K – Development and Template Developer

- Worked on the development environment setup.
- Developed dynamic templates using FastAPI and Jinja2.
- Worked on connecting the frontend templates with the backend.
- Supported application development and testing.

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	Yes (SQLAlchemy models, SQLite)
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL (if hosted)	NA (local deployment)
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Overview of the FitBuddy project's file and folder structure:

/FitBuddy	
/app	
main.py	- FastAPI application, routes, user-input handling
gemini_service.py	- Prompt creation and Gemini AI integration
database.py	- SQLAlchemy / SQLite setup
models.py	- Database models
/templates	- Jinja2 HTML templates (home, form, result, feedback)
/static	
/css	- Stylesheets
/js	- JavaScript files
/tests	- Test files
.env.example	- Environment configuration (GEMINI_API_KEY)
README.md	- Setup and run instructions
requirements.txt	- Python dependencies

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not hosted – runs locally at http://127.0.0.1:8000
Login Credentials (Demo)	NA – no login required (user login is a planned future enhancement)
Platform / Hosting Provider	Local deployment (Uvicorn server on localhost)
Repository Link	[Add your GitHub repository link]
Demo Video Link	[Add your demo video link]

Step 4: Run Instructions

Step-by-step instructions for the evaluator to run the project locally:

Pre-requisites: Python 3.10 or above, pip, Git, a web browser, an internet connection and a valid Google Gemini API key.

Step 1: Clone the repository: `git clone <repository-url>` and open the project folder (`cd FitBuddy`).

Step 2: (Optional) Create and activate a virtual environment: `python -m venv venv`, then `venv\Scripts\activate` (Windows) or `source venv/bin/activate` (Linux/Mac).

Step 3: Install dependencies: `pip install -r requirements.txt`

Step 4: Configure environment: copy `.env.example` to `.env` and add your Gemini API key (`GEMINI_API_KEY=your_key_here`).

Step 5: Start the application: `uvicorn app.main:app --reload` (the SQLite database is created automatically on first run).

Step 6: Open the browser at `http://127.0.0.1:8000`, click 'Start Fitness Journey', fill in the form and view the generated plan.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Requires an internet connection and a valid Gemini API key to generate plans.	Add a valid key in <code>.env</code> ; check connectivity if generation fails.
2	AI-generated plans may vary between runs and are not a substitute for professional medical or fitness advice.	Users with medical conditions or injuries should consult a qualified professional.
3	No user login, progress tracking or workout history yet; Gemini API rate limits may delay responses.	Planned as future enhancements; retry after a short wait.